

SF4: Data Logger – Interim Design Report

Smart Guitar Effects Processor & Data Logger

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1 Introduction & Application Specification

The primary objective of this project is to develop an advanced embedded system utilising the **Arduino Uno** hardware platform (powered by the ATmega328P MCU) configured as an interactive **Smart Guitar Effects Processor and Data Logger**. This directly addresses the requirement to specify an application demonstrating two-way communication between the MCU and PC.

2 Conceptual System Design & Communications

The system architecture is strictly partitioned into three specialised operational environments: **Analogue Circuitry**, **Arduino UNO R3 Firmware**, and **Host PC Software**.

Below is the block diagram for communications, illustrating the full signal path and JSON payload exchange over the 115200 Bd USB Serial Bridge.

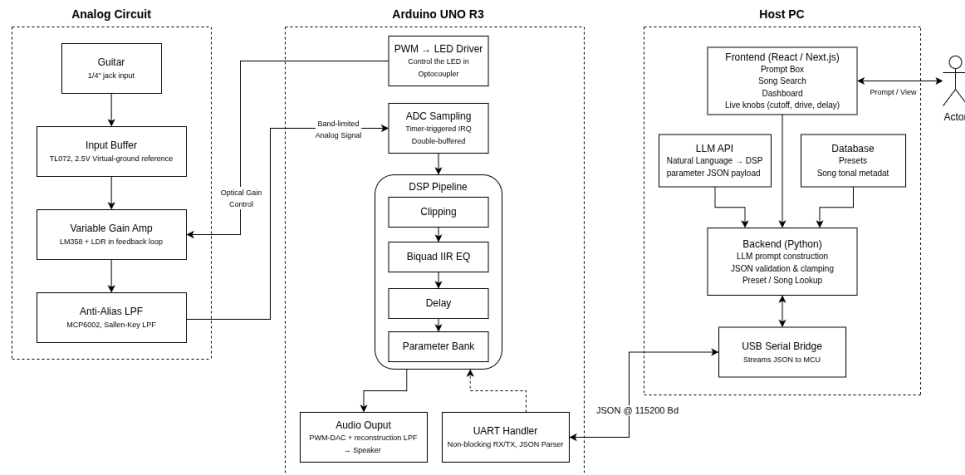


Figure 1: Systems and Communications Block Diagram.

3 Detailed Hardware Design

The custom analogue conditioning hardware consists of three distinct processing stages engineered to precisely scale and map a guitar’s low-voltage signal into the Arduino’s 0 – 5 V single-ended ADC input window. The complete analogue circuit design is provided below.

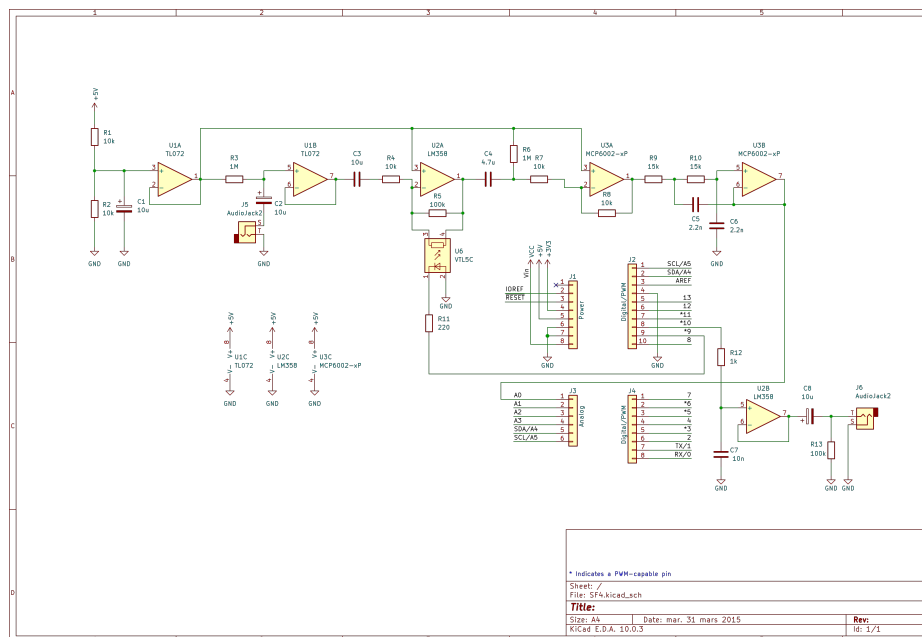


Figure 2: Analogue Circuit Design Schematic.

4 Firmware & Software Implementation Plan

The firmware architecture circumvents standard blocking Arduino function libraries to sustain low-latency real-time audio operations, utilising `Timer1` interrupts for 20 kHz sampling. The Host PC software utilises a React/Next.js frontend and Python backend to handle LLM-driven JSON parameter generation.

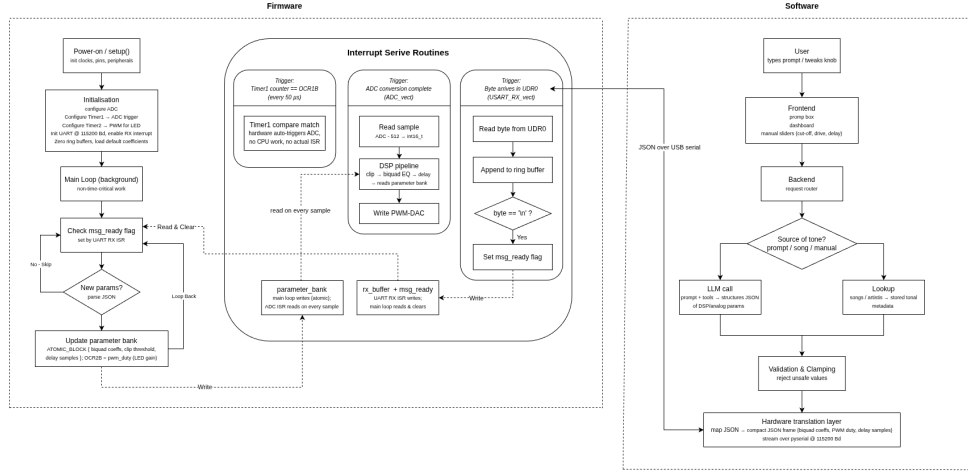


Figure 3: Firmware and Software Flow Diagram.

5 Component Requirements & Parts List

To assemble the target custom prototype circuit board, components have been selected within the strict **£15.00** project budget limit.

Table 1: Parts List for Purchasing (BOM)

Order Code	Brief Component Description	Qty	Unit Price (£)	Total (£)
SC17220	JFET-Input Dual Op-Amp (TL072CP)	2	1.000	2.000
SC17198	Dual Low-Power Op-Amps (LM358AP)	5 [†]	0.370	1.850
1292245	Rail-to-Rail Dual Op-Amp (MCP6002-I/P)	2	0.346	0.692
3692218	Optocoupler / Vactrol (NSL-32SR3)	1	3.290	3.290
CN23320	6.35mm (1/4") PCB Mono Jack Socket	1	0.930	0.930
CN22697	3.5mm Stereo Panel Jack Socket	1	0.330	0.330
Total Custom Expenditure (Budget: £15.00)				£9.092

[†] Component requires a minimum order quantity of 5 items.

6 Work Allocation

To optimise execution efficiency across the remaining two-week timeline, work tasks are divided as follows:

- **Gijeong Lee (CRSid: gl534)**: Primary ownership of hardware layout; responsible for breadboarding the analogue front-end, testing the Sallen-Key low-pass active filter blocks, and calibrating physical optocoupler compression metrics.
- **Jaehyung Choi (CRSid: jc2409)**: Primary ownership of software and firmware; responsible for creating register-direct `Timer1` interrupts, coding the real-time biquad filtering framework, configuring non-blocking asynchronous UART handlers, and building the HOST PC python/Next.js interface.